Periods of Dirac Spectral Sums on S^3

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Abstract

We determine the periods of spectral sums built from the Camporesi–Higuchi Dirac spectrum on the unit three-sphere, $|\lambda_n| = n+3/2$ with degeneracy $2(n+1)(n+2)$. (1) **T9:** the spectral zeta function of D^2 is a two-term polynomial in π^2 with rational coefficients at every integer $s \geq 1$; at half-integer $s \geq 5/2$ the same closed form carries odd zeta values (Theorem 1). (2) **Parity:** the Dirac Dirichlet series $D(s)$ lies in $\mathbb{Q}[\pi^2]$ at even s and is a rational combination of $\zeta(s-2)$ and $\zeta(s)$ at odd $s \geq 5$ (Theorem 2). (3) **The χ_{-4} split:** the sub-sums over even and odd n satisfy $D_{\text{even}}(s) - D_{\text{odd}}(s) = 2^{s-1}(\beta(s) - \beta(s-2))$ at every integer $s \geq 2$, so Catalan's G and $\beta(4)$ appear in each half at $s = 4$ and cancel in the sum (Theorem 3). (4) **Multi-loop chain sums:** the two-loop sum is $2S_{\min}$, with $S_{\min}$ in closed form in $\mathbb{Q}[\pi^2, \ln 2, \zeta(3), \zeta(5)]$ (Hoffman double t-values, each reducing to depth ≤ 1); the three-loop sum $S^{(3)} = 40.6554\ldots$ (rigorous bracket $[40.65339, 40.65578]$) has an explicit closed form in $\mathbb{Q}[\pi^2, \ln 2, \zeta(3), \zeta(5), \zeta(5,3)]$, in which $\zeta(5,3)$, at coefficient $6\pi^2$, is the only constant beyond single zeta values and every level-2 depth-2 generator cancels; the four-loop sum has realized depth ≤ 3. Realized depth is $\leq k-1$ at loop order $k = 2, 3, 4$ (Proposition 1). (5) **Sommerfeld sums:** the degeneracy-weighted coefficient sums D_p of the exact Dirac hydrogen spectrum are weight-$(2p-1)$ zeta polynomials in which $\zeta(2)\zeta(2p-3)$ cancels and $\max(0, \lfloor (2p-5)/2 \rfloor)$ products survive, observed through $p = 6$ (Observations 1, 2); a shell-sum identity removes $\zeta(s-1)$ from a fine-structure Dirichlet series (Theorem 5). The multi-loop sums are defined sums: their vertex weights come from a vertex model, the scalar-harmonic triangle with a parity clause (Eq. (6)). The physical vector-photon vertex on S^3 obeys a different rule (Paper 33), so the period statements are statements about the sums as defined. Two results whose reading depends on the vertex rule are stated for the model: the ground-state zero of the model's one-loop self-energy (Theorem 4), which fails for the physical vertex, and an unnormalized one-loop magnetic-moment ratio (Sec. 7), whose normalization is owed.

1 Introduction

On the unit three-sphere the Dirac operator has the explicit spectrum of Camporesi and Higuchi [7]. Sums over that spectrum — spectral zeta values, and the nested sums that arise when spinor levels are coupled through photon levels in a perturbative expansion — are explicit series whose values are periods. This paper determines which periods they are.

The questions are those of Paper 18's exchange-constant taxonomy [1]: which transcendentals enter, through which structure, and in which ring they sit. The answers here are exact. Single spectral sums are governed by operator order (Secs. 3–4) and by a split of the spectrum by the parity of n (Sec. 5). Nested sums over the level-2 (alternating) multiple zeta values have closed forms through three loops and a depth bound at four (Secs. 9–11). Sums over the exact Dirac hydrogen spectrum give Euler sums with a product-cancellation pattern (Sec. 12).

Scope. The nested sums use a vertex model (Sec. 2.3): the triangle rule for scalar harmonics with a parity clause, weighted by an $SO(4)$ channel count. They are defined sums, and every period statement below is a statement about them as defined. Whether the vertex model is the vertex of QED on S^3 is not a question this paper answers; the physical vector-photon vertex is computed in Paper 33, and it is not the model. Paper 33 also holds the finite-graph formulations of these diagrams. The two-term heat trace and the spectral action built on it are in Paper 51 [3]. All calculations use the `geovac` modules `qed_vertex`, `qed_self_energy`, `qed_anomalous_moment` and `qed_three_loop`; the backing tests are named at each result.

2 Spectral Data on S^3

2.1 Dirac spectrum

On the unit three-sphere, the Dirac operator has eigenvalues and degeneracies [7]

$$|\lambda_n| = n + \tfrac{3}{2}, \qquad g_n = 2(n+1)(n+2), \qquad n = 0, 1, 2, \ldots \tag{1}$$

The Dirac Dirichlet series is $D(s) = \sum_{n=0}^{\infty} g_n/|\lambda_n|^s$. Via Hurwitz zeta,

$$D(s) = 2\,\zeta(s{-}2, \tfrac{3}{2}) - \tfrac{1}{2}\,\zeta(s, \tfrac{3}{2}). \tag{2}$$

Using $\zeta(s, 3/2) = (2^s - 1)\zeta_R(s) - 2^s$, this becomes

$$D(s) = 2(2^{s-2} - 1)\,\zeta_R(s{-}2) - \tfrac{1}{2}(2^s - 1)\,\zeta_R(s). \tag{3}$$

We define the even-n and odd-n sub-sums:

$$D_{\text{even}}(s) = \sum_{\substack{n=0 \\ n\,\text{even}}}^{\infty} \frac{g_n}{|\lambda_n|^s}, \qquad D_{\text{odd}}(s) = \sum_{\substack{n=1 \\ n\,\text{odd}}}^{\infty} \frac{g_n}{|\lambda_n|^s}, \tag{4}$$

so that $D(s) = D_{\text{even}}(s) + D_{\text{odd}}(s)$.

2.2 Photon weights

The nested sums weight a photon level $q \geq 1$ by

$$\mu_q = q(q+2), \qquad d_q^T = q(q+2), \qquad q = 1, 2, 3, \ldots \tag{5}$$

These are the weights of the sums as defined, and the labelling of `geovac/hodge1_s3.py`. They are not the Hodge spectrum of the photon: the coclosed 1-forms on unit S^3 at level q have Hodge eigenvalue $(q+1)^2$, with $q(q+2)$ states in each of the two $\mathrm{SU}(2)_L \times \mathrm{SU}(2)_R$ components (Ikeda–Taniguchi 1978; Camporesi 1990; recomputed in Paper 33). The module's anchor test, `tests/test_hodge1_s3.py`, records the difference.

2.3 Vertex selection rule

The spectral sums of this paper couple spinor harmonics at levels n_1, n_2 to a photon mode at level q under the selection rule

$$|n_1 - n_2| \leq q \leq n_1 + n_2, \qquad n_1 + n_2 + q \text{ is odd.} \tag{6}$$

This is a *vertex model*, not the vertex of QED on S^3: the triangle bound is the one for scalar harmonics, and the parity clause is imposed. The physical vertex $\int_{S^3}\langle\psi_1, c(A)\,\psi_2\rangle$, with Clifford multiplication by a coclosed 1-form A, is computed exactly in Paper 33 [4]. It keeps the model's rule for same-sign spinors inside the model's window but also opens $q = n_1+n_2+1$, couples spinors of opposite sign of D at n_1+n_2+q even, and couples the ground state to itself at $q = 1$. Every sum below that uses Eq. (6) is a defined sum, and its period content is a property of that definition. The spectral-zeta results of Theorems 1–3 do not use the rule.

The SO(4) Clebsch–Gordan channel count $W(n_1, n_2, q)$ weights each allowed triple by the number of independent coupling channels (the two components of the transverse vector harmonic under $\mathrm{SU}(2)_L\times\mathrm{SU}(2)_R$). Inside the parity window the two spinors carry the same content $((n+1)/2, n/2)$, and both components pass the triangle and integrality conditions in both factors on every allowed triple, so $W = 2$ there and 0 elsewhere (Sec. 9.3).

3 Theorem 1: Squared Dirac Spectral Zeta (T9)

Theorem 1 (T9: D^2 spectral zeta is π^{even})**.** *The spectral zeta function of the squared Dirac operator on unit S^3 satisfies*

$$\zeta_{D^2}(s) = 2^{2s-1}\big[\lambda(2s{-}2) - \lambda(2s)\big], \tag{7}$$

where $\lambda(2k) = (1-2^{-2k})\,\zeta_R(2k) = \text{rational}\times\pi^{2k}$. At every integer $s \geq 1$, $\zeta_{D^2}(s)$ is a two-term polynomial in π^2 with rational coefficients; no odd-zeta content appears at any integer $s \geq 1$. The statement is an integer-s one: at half-integer $s \geq 5/2$ the same formula (7) *carries odd zeta values (e.g. $\zeta_{D^2}(5/2) = 14\,\zeta(3) - \frac{31}{2}\,\zeta(5)$), and $s = 1/2, 3/2$ are simple poles with rational residues.*

Proof. The squared Dirac eigenvalues are $\lambda_n^2 = (n+3/2)^2 = (2n+3)^2/4$, so $\zeta_{D^2}(s) = \sum_{n\geq 0} g_n \cdot (4/(2n+3)^2)^s = 4^s\sum_{n\geq 0} 2(n+1)(n+2)/(2n+3)^{2s}$. Substituting $m = 2n+3$ (which runs over odd integers ≥ 3), the sum becomes $\sum_{m\geq 3, m\,\text{odd}}(m^2-1)/2\cdot 4^s/m^{2s}$. The $(m^2-1)/2$ factor splits as $(m^{2-2s} - m^{-2s})/2$, giving two Dirichlet-type sums over odd integers. Each odd-integer sum is $\lambda(2k) = (1-2^{-2k})\zeta_R(2k)$ with the first three odd values ($m = 1$) added or subtracted to correct the lower limit. By the Euler–Bernoulli formula, $\zeta_R(2k) = (-1)^{k+1}(2\pi)^{2k}B_{2k}/(2\cdot(2k)!)$, so $\lambda(2k) = \text{rational}\times\pi^{2k}$. At integer s both arguments $2s-2$ and $2s$ are even, so odd-zeta values $\zeta_R(2k+1)$ cannot appear; at half-integer $s \geq 5/2$ both arguments are odd and ≥ 3, so $\lambda(2s-2)$, $\lambda(2s)$ are rational multiples of odd zeta values (at $s = 1/2, 3/2$ an argument equals 1, a pole; at negative half-integer s the values $\zeta(1-2k) = -B_{2k}/(2k)$ are rational). □

Verification: Symbolic identity confirmed in `sympy` for $s = 1, \ldots, 10$. Numerical agreement to 10^{-60} at $s = 2, \ldots, 20$. The integer-s closed forms at $s = 2, 3$ and the half-integer value $\zeta_{D^2}(5/2) = 14\,\zeta(3) - \frac{31}{2}\,\zeta(5)$ are checked against the direct spectral sum in `tests/test_paper28_t9_half_integer.py`.

Table 1: Squared Dirac spectral zeta $\zeta_{D^2}(s)$.

s	$\zeta_{D^2}(s)$
1	$-\pi^2/4$
2	$\pi^2 - \pi^4/12$
3	$\pi^4/3 - \pi^6/30$
4	$2\pi^6/15 - 17\pi^8/1260$

Remark 1 (Where $\sqrt{\pi}$ lives: heat-trace coefficients, not ζ_{D^2})**.** *The $\sqrt{\pi}$ met in heat-kernel-side analyses of ζ_{D^2} does not live in the spectral zeta function at any s. It lives only in the heat-trace coefficients: the short-time expansion $\operatorname{Tr} e^{-tD^2} \sim \frac{\sqrt{\pi}}{2}\, t^{-3/2} - \frac{\sqrt{\pi}}{4}\, t^{-1/2}$ carries the $(4\pi t)^{-3/2}$-type prefactor of an odd-dimensional heat kernel, and the Mellin transform $\zeta_{D^2}(s) = \Gamma(s)^{-1} \int_0^\infty t^{s-1} \operatorname{Tr} e^{-tD^2}\, dt$ divides each coefficient by $\Gamma(s)$ at its pole. At $s = 3/2$ this gives the residue $(\sqrt{\pi}/2)/\Gamma(3/2) = 1$, and at $s = 1/2$ the residue $(-\sqrt{\pi}/4)/\Gamma(1/2) = -1/4$: both rational. Away from the poles, the values at half-integer $s \geq 5/2$ are odd-zeta combinations (Theorem 1), not multiples of $\sqrt{\pi}$. At integer $s \geq 1$ the values are pure $\pi^{2k} \cdot \mathbb{Q}$ (Table 1 above). After the standard Connes–Chamseddine volume normalization (multiplication of the heat-trace coefficients by $(4\pi)^{d/2}$ with $d = 3$), the coefficients also cleanly enter $\mathbb{Q}[\pi^2]$: Paper 51 [3] Cor. 2.1 gives $a_0^{D^2} = 4\pi^2$, $a_1^{D^2} = -2\pi^2$, $a_k = 0$ for $k \geq 2$. At integer s the GeoVac S^3 M2 sub-mechanism is therefore mixed-Tate over $\mathbb{Q}$ in the F–M sense (Paper 32 [2] § VIII Corollary), in the strictly smaller pure-Tate sub-ring $\bigoplus_k \pi^{2k} \cdot \mathbb{Q}$.*

4 Theorem 2: Even/Odd Parity Discriminant

Theorem 2 (Parity discriminant)**.** *At even integer s, the Dirac Dirichlet series $D(s)$ is purely π^{even}. At odd integer s, $D(s)$ involves $\zeta_R(s-2)$ and $\zeta_R(s)$ (odd-zeta values; $s \geq 5$, since $\zeta_R(1)$ is a pole). Operator order (2nd vs 1st) is the primary transcendental discriminant, conditional on the period conjecture $\zeta(2k+1) \notin \mathbb{Q}(\pi)$: the second-order operators Δ_{LB} and D^2 produce π^{even} exclusively at integer s (this holds for Δ_{LB} and D^2, with spectra $n^2 - 1$ and $(n + 3/2)^2$, whose partial fractions have rational poles; it is not a property of every second-order operator — the spinor connection Laplacian $\nabla^*\nabla = D^2 - 3/2$ has $\zeta_{\nabla^*\nabla}(2) = \frac{1}{12}\big(14\,\pi \tan(\frac{\pi\sqrt{6}}{2})/\sqrt{6} + 5\pi^2 \sec^2(\frac{\pi\sqrt{6}}{2})\big)$, outside $\mathbb{Q}[\pi^2]$) (their spectral determinants are not covered: $\zeta'_{D^2}(0) = 3\zeta_R(3)/(4\pi^2) + \frac{1}{2}\ln 2$); first-order operators produce odd-zeta values at odd s.*

Proof. From Eq. (3), $D(s) = 2(2^{s-2} - 1)\,\zeta_R(s-2) - \frac{1}{2}(2^s - 1)\,\zeta_R(s)$. At even s, both $\zeta_R(s-2)$ and $\zeta_R(s)$ are Bernoulli-type: $\zeta_R(2k) = (-1)^{k+1}(2\pi)^{2k} B_{2k}/(2 \cdot (2k)!) = \text{rational} \times \pi^{2k}$. At odd s, both $\zeta_R(s-2)$ and $\zeta_R(s)$ are genuine odd-zeta values ($\zeta(3)$, $\zeta(5)$, …), conjecturally $\mathbb{Q}$-linearly independent of all π^{2k} (open: Apéry's theorem gives the irrationality of $\zeta(3)$ and Rivoal's infinitely many irrational odd zeta values [18, 19]; neither gives independence from π). □

Table 2: Dirac Dirichlet series $D(s)$: PSLQ identifications at 80 dps.

s	$D(s)$	Class
4	$\pi^2 - \pi^4/12$	π^{even}
5	$14\,\zeta(3) - \frac{31}{2}\,\zeta(5)$	odd-zeta
6	$\pi^4/3 - \pi^6/30$	π^{even}
7	$62\,\zeta(5) - \frac{127}{2}\,\zeta(7)$	odd-zeta
8	$2\pi^6/15 - 17\pi^8/1260$	π^{even}
9	$254\,\zeta(7) - \frac{511}{2}\,\zeta(9)$	odd-zeta
10	$17\pi^8/315 - 31\pi^{10}/5670$	π^{even}

5 Theorem 3: The χ_{-4} Identity

Theorem 3 (χ_{-4} identity)**.** *For every integer $s \geq 2$ (the even/odd spectral sums converge for $s \geq 4$; at $s = 2, 3$ by Hurwitz continuation),*

$$D_{\text{even}}(s) - D_{\text{odd}}(s) \;=\; 2^{s-1}\big(\beta(s) - \beta(s{-}2)\big), \tag{8}$$

where $\beta(s) = L(s, \chi_{-4})$ is the Dirichlet beta function.

Proof. Express $D_{\text{even}}(s)$ and $D_{\text{odd}}(s)$ via Hurwitz zeta at quarter-integer shifts. With $m = n + 3/2$ and full degeneracy $g_n = 2(n+1)(n+2) = 2(m^2 - 1/4)$, the even-$n$ eigenvalues are $m = 2(k + 3/4)$ and the odd-n ones $m = 2(k + 5/4)$, $k \geq 0$, so $D_{\text{even}}(s) = 2^{-(s-3)}\zeta(s{-}2, 3/4) - 2^{-(s+1)}\zeta(s, 3/4)$ and $D_{\text{odd}}(s) = 2^{-(s-3)}\zeta(s{-}2, 5/4) - 2^{-(s+1)}\zeta(s, 5/4)$. Two elementary Hurwitz identities: $\zeta(s, 5/4) = \zeta(s, 1/4) - 4^s$ (shift recurrence) and $\zeta(s, 3/4) - \zeta(s, 1/4) = -4^s\beta(s)$ (the Dirichlet beta definition). Together, $\zeta(s, 3/4) - \zeta(s, 5/4) = 4^s(1 - \beta(s))$; substituting into the difference, $D_{\text{even}} - D_{\text{odd}} = 2^{s-1}(1{-}\beta(s{-}2)) - 2^{s-1}(1{-}\beta(s)) = 2^{s-1}(\beta(s){-}\beta(s{-}2))$, which is (8). The Hurwitz decompositions and the closed form are verified bit-exactly against direct spectral sums at $s \in \{4, 5, 6, 7\}$ to 40 digits. □

Verification: Eq. (8) verified symbolically (`sympy` residual zero) at $s \in \{2, 3, \ldots, 10, 15\}$ and by PSLQ at 100-digit precision at every integer s in the tested range. At $s = 4$, $D_{\text{even}}(4) - D_{\text{odd}}(4) = 8(\beta(4) - G)$ (since $G = \beta(2)$), consistent with the explicit decompositions (9)–(10) below.

5.1 The even/odd split at $s = 4$

The even-n and odd-n sub-sums of $D(4)$ decompose as:

$$D_{\text{even}}(4) = \tfrac{\pi^2}{2} - \tfrac{\pi^4}{24} - 4G + 4\beta(4), \tag{9}$$

$$D_{\text{odd}}(4) = \tfrac{\pi^2}{2} - \tfrac{\pi^4}{24} + 4G - 4\beta(4), \tag{10}$$

$$D(4) = \pi^2 - \tfrac{\pi^4}{12} \qquad \text{(Catalan and } \beta(4) \text{ cancel).} \tag{11}$$

Here $G = \beta(2) = \sum_{n=0}^{\infty}(-1)^n/(2n+1)^2 \approx 0.9159656$ is Catalan's constant and $\beta(4) \approx 0.9889446$ is the Dirichlet beta function at $s = 4$. The mechanism is the quarter-integer Hurwitz shift: restricting n to one parity class puts the eigenvalues $m = n + 3/2$ on $2(k + 3/4)$ or $2(k + 5/4)$, and $L(s, \chi_{-4})$ appears in the difference. In the vertex model of Sec. 2.3 a parity restriction of this kind is what the parity clause imposes on internal lines; that reading belongs to the model. Verified by PSLQ at 80-digit precision with basis $\{1, \pi^2, \pi^4, G, \beta(4)\}$ (`tests/test_smin_chi_neg4.py`, `tests/test_qed_vertex.py`).

Remark 2 (Motivic home of Theorem 3)**.** *Each of $D_{\text{even}}(s)$, $D_{\text{odd}}(s)$, and their difference is a period of mixed Tate motives at level 4 over $\mathcal{O}_4[1/4] = \mathbb{Z}[i, 1/2]$, by direct application of Deligne's classification at $N = 4$ and the explicit basis of Glanois (Paper 32 § VIII Corollary `cor:m3_cyclotomic_mixed_tate`; Paper 18 § III.7; `tests/test_paper55_m3_cyclotomic_mixed_tate.py`). The split can be read as a Galois descent from level 4 to level 2: the un-restricted $D(s) = D_{\text{even}} + D_{\text{odd}}$ lives in the smaller level-≤ 2 ring (Theorem 1 and Theorem 2); the $\zeta(s, 1/4)$ pieces in $D_{\text{even}}, D_{\text{odd}}$ individually cancel in the sum but survive in the difference, exposing the level-4 Dirichlet L-value content. Catalan $G = \beta(2)$ and $\beta(4)$ are periods in $\mathcal{MT}(\mathbb{Z}[i, 1/2])$ at depth 1 but conjecturally not in $\mathcal{MT}(\mathbb{Z}[1/2])$ at any depth, so the level-4 content is carried by the parity split.*

5.2 Three axes

The single spectral sums of Secs. 3–5 are sorted by three independent axes:

1. **Operator order.** Second-order (D^2) $\to \pi^{\text{even}}$ at integer s (Theorem 1); first-order ($|D|$) $\to$ odd-zeta at odd $s \geq 5$ (Theorem 2).
2. **Bundle type.** Scalar vs spinor modulates coefficients but not the transcendental class (Paper 18 Sec IV).
3. **Restriction of the spectrum.** Restricting n to one parity class introduces the Dirichlet L-values G, $\beta(4)$, absent from the unrestricted sums (Theorem 3).

6 Theorem 4: Ground-State Zero of the Model Self-Energy

The model's one-loop self-energy sum is

$$\Sigma(n_{\text{ext}}) = \sum_{n_{\text{int}},q} \frac{W(n_{\text{ext}}, n_{\text{int}}, q)\; g_{n_{\text{int}}}\; d_q^T}{|\lambda_{n_{\text{int}}}|^{2s_e}\; \mu_q^{s_\gamma}}, \tag{12}$$

where the sum is over internal electron mode n_{int} and photon mode q subject to the vertex selection rule (6), $s_e = 2$ (so $|\lambda|^4$ in the denominator), $s_\gamma = 1$, and W is the SO(4) channel count.

Theorem 4 (Self-energy zero of the vertex model)**.** *Under the selection rule* (6)*,* $\Sigma(0) = 0$ *exactly: the model's one-loop self-energy vanishes identically at the Dirac ground state* $n_{\text{ext}} = 0$*.*

Proof. At $n_{\text{ext}} = 0$, the vertex selection rule requires $|0 - n_{\text{int}}| \leq q \leq 0 + n_{\text{int}}$ and $0 + n_{\text{int}} + q$ odd. The first condition forces $q = n_{\text{int}}$. The second then requires $2\,n_{\text{int}}$ to be odd, which is impossible for any integer n_{int}. Therefore $W(0, n_{\text{int}}, q) = 0$ for all (n_{int}, q), and the sum is empty. □

The theorem is a statement about the model; it is backed by `tests/test_qed_self_energy.py`. The zero comes from the model's triangle bound, which is the scalar one. It fails for the physical vector-photon vertex: there the ground state couples at $q = n_{\text{int}} + 1$ to same-sign internal levels and at $q = n_{\text{int}}$ to opposite-sign ones, so the corresponding sum is a sum of positive terms with terms present (Paper 33).

6.1 Self-energy for excited states

For $n_{\text{ext}} \geq 1$, the model self-energy is nonzero and convergent at $s_e = 2$, $s_\gamma = 1$, and increases monotonically with n_{ext}.

6.2 Vertex correction

The one-loop vertex correction involves two vertex insertions:

$$\Lambda(n_{\text{ext}}) = \sum_{n_{\text{int}},q_1,q_2} \frac{W(n_{\text{ext}}, n_{\text{int}}, q_1)\; W(n_{\text{int}}, n_{\text{ext}}, q_2)\; g_{n_{\text{int}}}\; d_{q_1}^T\; d_{q_2}^T}{|\lambda_{n_{\text{int}}}|^{2s_e}\; \mu_{q_1}^{s_\gamma}\; \mu_{q_2}^{s_\gamma}}. \tag{13}$$

This is $O(N^3)$ in the cutoff. Both $\Lambda(n_{\text{ext}})$ and $\Sigma(n_{\text{ext}})$ are backed by 35 tests in `tests/test_qed_self_energy.py`. **Note:** The shell-summed vertex correction $\Lambda(n_{\text{ext}})$ defined above uses the CG-weight channel count W and is m_j-independent (it gives $B = 0$ by the Wigner–Eckart theorem). The anomalous magnetic moment extraction requires the m_j-resolved three-point function developed in the next section.

7 A One-Loop Magnetic-Moment Ratio in the Vertex Model

This section is a statement about the vertex model. Its sums carry unit coupling, no loop measure and the photon weights of Eq. (5), and the loop vertex obeys Eq. (6). Normalizing them — the coupling, the transverse photon propagator with its mode measure and the Hodge eigenvalue $(q+1)^2$, and the physical vertex of Paper 33 — is owed work, so no number below is an anomalous moment. The algebraic content (Clebsch–Gordan sums over rational denominators) is what the section records.

7.1 Why the two-point vertex correction gives $B = 0$

The two-point vertex correction $\Lambda(n_{\text{ext}})$ defined in Eq. (13) sums over all internal modes and photon polarizations at *fixed* external quantum numbers $(n_{\text{ext}}, j_{\text{ext}}, m_j)$. On S^3, the vertex amplitude at each junction involves a product of four $\text{SU}(2)_L \times \text{SU}(2)_R$ Clebsch–Gordan coefficients. When the photon polarizations $(m_{g,L}, m_{g,R})$ are summed over at both vertices, the SO(4) Wigner–Eckart theorem forces the result to be independent of m_j:

$$\Lambda(n_{\text{ext}}, j, m_j = +\tfrac{1}{2}) = \Lambda(n_{\text{ext}}, j, m_j = -\tfrac{1}{2}). \tag{14}$$

Therefore the spin-dependent piece $B_{2\text{pt}} = \Lambda(+\frac{1}{2}) - \Lambda(-\frac{1}{2}) = 0$ identically. The two-point vertex correction is a scalar under SO(4) rotations and cannot distinguish spin orientations. To extract the anomalous magnetic moment (F_2 form factor), an external probe must break the rotational symmetry.

7.2 Three-point vertex correction with probe photon

The anomalous magnetic moment is extracted from a three-point function in which an external probe photon of mode $q_{\text{probe}} = 1$ (the lowest transverse vector harmonic, the S^3 analog of a uniform magnetic field) is inserted between the two loop-photon vertices. The topology is:

$$\langle \text{ext}, m_j |\, \Gamma^{1-\text{loop}}_{\text{probe}}\, | \text{ext}, m_j \rangle = \sum_{\substack{n_{\text{int}},\, j_{\text{int}} \\ m_{j,\text{int}},\, m'_{j,\text{int}} \\ q_{\text{loop}}}} \frac{V_1 \;\cdot\; V_{\text{probe}} \;\cdot\; V_3}{|\lambda_{n_{\text{int}}}|^4 \; \mu_{q_{\text{loop}}}}, \tag{15}$$

where:

- V_1: amplitude for the external electron (m_j) to emit a loop photon and transition to the internal state $(m_{j,\text{int}})$;
- V_{probe}: amplitude for the probe photon at $q = 1$ to scatter the internal electron from $m_{j,\text{int}}$ to $m'_{j,\text{int}}$, summed over probe polarizations;
- V_3: amplitude for the internal state $(m'_{j,\text{int}})$ to reabsorb the loop photon and return to the external state (m_j).

Each vertex amplitude is a product of four Clebsch–Gordan coefficients coupling the $\text{SU}(2)_L \times \text{SU}(2)_R$ representations of source, target, and photon:

$$V = \sum_{m_{L,1}} C^{j_s}_{j_{s,L}\, j_{s,R}}(m_{L,1},\, m_j - m_{L,1};\, m_j) \cdot C^{j_t}_{j_{t,L}\, j_{t,R}}(m_{L,2},\, m_{R,2};\, m'_j) \cdot C^{j_{t,L}}_{j_{s,L}\, j_{g,L}}(m_{L,1},\, m_{g,L};\, m_{L,2}) \cdot C^{j_{t,R}}_{j_{s,R}\, j_{g,R}}(m_{R,1},\, m \tag{16}$$

where $m_{L,2} = m_{L,1} + m_{g,L}$ and $m_{R,2} = m_{R,1} + m_{g,R}$, and the sum runs over all $m_{L,1} \in \{-j_{s,L}, \ldots, +j_{s,L}\}$.

The probe photon at $q = 1$ has channels $(j_{g,L}, j_{g,R}) = (1, 0)$ or $(0, 1)$, corresponding to the two components of the transverse vector harmonic. The probe couples internal states $m_{j,\mathrm{int}} \to m'_{j,\mathrm{int}}$, enabling spin-dependent amplitudes that the polarization-summed two-point function cannot access.

The spin-dependent piece of the vertex correction is:

$$B = \Gamma^{1-\mathrm{loop}}_{\mathrm{probe}}(m_j = +\tfrac{1}{2}) - \Gamma^{1-\mathrm{loop}}_{\mathrm{probe}}(m_j = -\tfrac{1}{2}). \tag{17}$$

7.3 Tree-level normalization

The tree-level (zeroth-order) coupling of the external electron to the $q = 1$ probe photon provides the normalization for F_2 extraction. At the external state $n_{\mathrm{ext}} = 1$, $j = 1/2$ (Camporesi–Higuchi level $n_{\mathrm{CH}} = 1$, with $j_L = 1$, $j_R = 1/2$):

$$V_{\mathrm{tree}}(m_j = +\tfrac{1}{2}) = -V_{\mathrm{tree}}(m_j = -\tfrac{1}{2}) = 0.2790, \tag{18}$$

$$V_{\mathrm{magnetic}} = V_{\mathrm{tree}}(+\tfrac{1}{2}) - V_{\mathrm{tree}}(-\tfrac{1}{2}) = 0.5579. \tag{19}$$

The charge piece $[V_{\mathrm{tree}}(+\frac{1}{2}) + V_{\mathrm{tree}}(-\frac{1}{2})]/2 = 0$ vanishes—the $q = 1$ probe photon couples purely to the magnetic moment at tree level, as expected for the S^3 analog of a uniform magnetic field.

Analytically, $V_{\mathrm{magnetic}} = 2(\sqrt{2}/3 - \sqrt{3}/9)$, a pure algebraic number from the Clebsch–Gordan coefficients.

The Pauli form factor is then:

$$F_2 = \frac{B}{V_{\mathrm{magnetic}}}, \tag{20}$$

and the anomalous magnetic moment on S^3 is $a_e^{S^3} = F_2$.

7.4 Convergence and results

The three-point vertex correction converges rapidly. Table 3 shows the per-level contributions for the external state at $n_{\mathrm{ext}} = 1$.

Table 3: Per-level contributions to the anomalous magnetic moment on S^3. External state $n_{\mathrm{ext}} = 1$, $j = 1/2$. The $n_{\mathrm{int}} = 0$ contribution vanishes in the vertex model (Theorem 4); with the vector-photon rule the $q = 1$ channel there is open. Contributions drop by approximately $100\times$ per successive allowed level pair.

n_{int}	ΔB	B_{cumul}	$F_2/[\alpha/(2\pi)]$	Status
0	0	0	—	closed in the vertex model
1	6.96×10^{-4}	6.96×10^{-4}	1.074	99% of final
2	3.76×10^{-6}	7.00×10^{-4}	1.080	
3	2.85×10^{-6}	7.02×10^{-4}	1.084	
4	1.34×10^{-7}	7.025×10^{-4}	1.084	
5	1.25×10^{-7}	7.027×10^{-4}	1.084	
6	1.25×10^{-8}	7.027×10^{-4}	1.084	
7	1.36×10^{-8}	7.027×10^{-4}	1.084	
8	2.00×10^{-9}	7.027×10^{-4}	1.084	

Three structural features are visible:

1. **Rapid convergence.** The $n_{\text{int}} = 1$ level contributes 99% of the final value. Each subsequent allowed level pair contributes roughly 100× less than the previous one.

2. **Vertex parity filtering (model).** The $n_{\text{int}} = 0$ contribution vanishes by the model rule (6) (the loop-photon vertex at $n_{\text{ext}} = 1$, $n_{\text{int}} = 0$ requires $q_{\text{loop}} = 1$ and $1 + 0 + 1 = 2$ is even, forbidden). The dominant contribution comes from $n_{\text{int}} = 1$ (the nearest allowed level to the external state). Higher even-n_{int} levels contribute but are suppressed relative to odd-n_{int} neighbors.

3. **The ratio.** The ratio B/V_{mag} divided by the CODATA value of $\alpha/(2\pi)$ converges to 1.084.

7.5 The ratio at $n_{\text{ext}} = 1$

At $n_{\max} = 12$ (converged to 5 significant figures; the value is already stable at $n_{\max} = 8$):

$$\frac{F_2}{\alpha/(2\pi)} = 1.084, \tag{21}$$

with $B = 7.027 \times 10^{-4}$ and $V_{\text{magnetic}} = 0.5579$.

Normalization [OBSERVATION]. B and V_{mag} are sums of Clebsch–Gordan products over spectral denominators with unit coupling: neither carries e^2, a loop measure or the harmonic normalization on S^3. The ratio $F_2 = B/V_{\text{mag}} = 1.2596 \times 10^{-3}$ is therefore not an anomalous moment in units of α, and Eq. (21) divides it by the CODATA value of $\alpha/(2\pi)$; the coupling does not cancel. The value 1.084 is a numerical observation about this unnormalized ratio, computed moreover in the vertex model (the $n_{\text{int}} = 0$, $q = 1$ channel that the physical vertex opens is closed). The flat-space limit $R \to \infty$ is not computed.

7.6 Mode dependence

Table 4 shows the same ratio at $n_{\text{ext}} = 1, \ldots, 5$.

Table 4: The model's magnetic-moment ratio on unit S^3 by external level. Each row reports the tree-level magnetic coupling V_{mag}, the one-loop correction B, and B/V_{mag} divided by the CODATA $\alpha/(2\pi)$ (unnormalized; see above). The $n_{\text{ext}} = 1$ result uses $n_{\max} = 12$ (converged to 5 digits); $n_{\text{ext}} = 2$–5 use $n_{\max} = 5$ (converged to 3 digits via n^{-7} power-law tail). Exact algebraic forms for V_{mag} and the dominant $B(n_{\text{int}} = 1)$ are available at each n_{ext}; all expressions are sums of square roots with rational coefficients (no transcendentals).

n_{ext}	λ_{ext}	V_{mag}	B	ratio to CODATA $\alpha/(2\pi)$	ratio -1
1	5/2	0.5579	7.027×10^{-4}	1.084	+8.4%
2	7/2	0.3893	2.772×10^{-4}	0.613	−39%
3	9/2	0.3001	1.296×10^{-4}	0.370	−63%
4	11/2	0.2445	6.940×10^{-5}	0.244	−76%
5	13/2	0.2064	4.139×10^{-5}	0.173	−83%

The loop correction B drops faster with n_{ext} than the tree-level coupling: across the five levels B spans a factor of 17× while V_{mag} spans 2.7×. The tree-level amplitude follows $V_{\text{mag}} \to (4/3)/|\lambda|$ asymptotically (Richardson-extrapolated to 4-digit agreement with 4/3). A leading curvature form $1 + R_{\text{scalar}}/(12|\lambda|^2)$ (motivated by the curved-space formalism of [6]) gives 1.080 at $n_{\text{ext}} = 1$; given

the normalization above, the closeness to 1.084 is a numerical coincidence, and a two-term fit in $1/|\lambda|^2$ to the five levels gives a leading coefficient near -20, not $1/2$. With one curvature value per level, coefficients beyond the first cannot be separated.

The five computed values are well described by a power law:

$$\frac{F_2}{\alpha/(2\pi)} = (5.91 \pm 0.36)\ |\lambda|^{-1.84\pm0.06}. \tag{22}$$

The fitted exponent is 2.8σ from -2; the forced $p = -2$ fit has twice the RMS residual. Analysis of the dominant internal-mode contribution ($n_{\rm int} = 1$) shows its local log-log slope converging toward -2 ($-2.57 \to -2.26 \to -2.16 \to -2.11$ across $n_{\rm ext} = 1, \ldots, 5$), but higher internal modes contribute an increasing fraction at larger $n_{\rm ext}$ (99% at $n_{\rm ext} = 1$ vs. 69% at $n_{\rm ext} = 3$), pulling the effective exponent above -2. Whether the true asymptotic exponent is exactly -2 (with finite-curvature corrections) or a distinct value remains an open question.

The flat-space limit is $R \to \infty$ at fixed physical mass ($n \sim mR \to \infty$ simultaneously), not $|\lambda| \to \infty$ on a fixed-size sphere, and it is not computed here.

Verification: The computation uses exact `sympy` Clebsch–Gordan coefficients at every vertex, with no numerical approximation in the angular algebra. The propagator denominators are exact rationals ($|\lambda_n|^4$ and μ_q from Eqs. (1) and (5)). Backing tests: `tests/test_qed_anomalous_moment.py`.

8 Heat Trace and Spectral Action

The Dirac heat trace on unit S^3 is two-term up to exponentially small corrections,

$$\operatorname{Tr} e^{-tD^2} = \frac{\sqrt{\pi}}{2}\, t^{-3/2} - \frac{\sqrt{\pi}}{4}\, t^{-1/2} + O(e^{-\pi^2/t}), \tag{23}$$

a result of Chamseddine and Connes [20]. Equivalently $\zeta_{D^2}(-k) = 0$ for every integer $k \geq 0$, the negative-integer companion of Theorem 1. The proof by Bernoulli polynomials at $3/2$, the spectral action on S^3_R and on $S^3 \times S^1$ that follows from it, the scalar and transverse-traceless comparison heat traces, and the graviton diagnostics on the truncated spectrum are in Paper 51 [3]. The cutoff Λ defined by $\operatorname{Tr} e^{-D^2/\Lambda^2} = K/\pi$, with K the number of Paper 2's Observation, and the integer-relation search on it, are in Paper 2's table of examined mechanisms [5].

9 Two-Loop Identified Constant: $S_{\rm min}$

The model's CG-weighted two-loop sunset sum is $2S_{\rm min}$ (Sec. 9.4), with

$$S_{\rm min} = \sum_{k=1}^{\infty} T(k)^2, \qquad T(k) = 2\,\zeta(2, k+\tfrac{3}{2}) - \tfrac{1}{2}\,\zeta(4, k+\tfrac{3}{2}). \tag{24}$$

This is a depth-2 multiple Hurwitz zeta value at half-integer shift $a = 3/2$, a sum over the Dirac spectrum alone; the vertex rule enters only through the relation $S_{\rm sunset} = 2S_{\rm min}$.

9.1 200-digit computation

$S_{\rm min}$ is computed to 200 verified digits using `mpmath.nsum` with Levin u-transform acceleration. The first 80 digits agree with three independent methods (direct summation + explicit tail; Levin u-transform; Euler–Maclaurin 26-term asymptotic):

$$S_{\rm min} = 2.47993\,69380\,34222\,55441\,35795\,00082\,93821\,44688\ldots \tag{25}$$

9.2 Motivic identification: Hoffman double t-values

$S_{\min}$ decomposes *exactly* into Hoffman double t-values — classical level-2 alternating MZVs, periods of $\mathcal{MT}(\mathbb{Z}[1/2])$. Notation: $\lambda(s) = (1-2^{-s})\zeta(s) = \sum_{o\geq 1,\,\text{odd}} o^{-s}$; $t(b,c) = \sum_{o_2>o_1\geq 1,\,\text{odd}} o_2^{-b} o_1^{-c}$ (Hoffman [11] descending convention: first index b on the larger summation variable).

8-t-value decomposition (verified to residual 1.8×10^{-129} at 130 dps, re-verified at 260 dps):

$$\begin{aligned} S_{\min} = {} & 64\big[t(2,1) - t(4,1) - 3\,t(2,2) + 3\,t(4,2) - t(2,3) + t(4,3) + 3\,t(2,4) - 3\,t(4,4)\big] \\ & + 32\sum_{s=3}^{8} d_s\left(\lambda(s) - 1 - 3^{-s}\right), \end{aligned} \tag{26}$$

where the diagonal coefficients are $d_3 = 1$, $d_4 = -3$, $d_5 = -2$, $d_6 = 6$, $d_7 = 1$, $d_8 = -3$.

Stuffle reductions (exact algebraic identities from the ordering decomposition $\lambda(b)\lambda(c) = t(b,c) + t(c,b) + \lambda(b+c)$):

$$t(2,2) = \tfrac{1}{2}\big(\lambda(2)^2 - \lambda(4)\big), \tag{27}$$

$$t(4,4) = \tfrac{1}{2}\big(\lambda(4)^2 - \lambda(8)\big), \tag{28}$$

$$t(2,4) + t(4,2) = \lambda(2)\lambda(4) - \lambda(6). \tag{29}$$

(The antisymmetric combination $t(2,4) - t(4,2)$ enters $S_{\min}$ with coefficient zero, so only the reducible symmetric combination appears.)

Odd-weight depth-1 reductions (verified at $\geq 1\times 10^{-130}$; both are verbatim rows of Hoffman [11] Appendix A):

$$t(2,1) = -\tfrac{7}{16}\,\zeta(3) + \tfrac{1}{8}\,\pi^2\ln 2, \tag{30}$$

$$t(2,3) = -\tfrac{31}{64}\,\zeta(5) + \tfrac{1}{16}\,\pi^2\,\zeta(3). \tag{31}$$

Remaining odd-weight components — identified. PSLQ at 220 dps against the full conjectural level-2 bases (dim 8 at weight 5; 19 product monomials at weight 7) identifies both remaining components (residuals 2.8×10^{-221} and 3.6×10^{-222}):

$$t(4,1) = \tfrac{1}{96}\,\pi^4\ln 2 - \tfrac{1}{64}\,\pi^2\zeta(3) - \tfrac{31}{64}\,\zeta(5), \tag{32}$$

$$t(4,3) = \tfrac{1}{128}\,\pi^4\zeta(3) - \tfrac{5}{128}\,\pi^2\zeta(5) - \tfrac{127}{256}\,\zeta(7). \tag{33}$$

Both identified forms are likewise verbatim rows of Hoffman [11] Appendix A ($t(4,1)$ directly; $t(4,3)$ after converting his t-basis) — the PSLQ runs independently rediscovered the published table (grounding verified against the source PDF, 2026-06-11). Assembling the full decomposition yields the explicit closed form

$$S_{\min} = 8\pi^2\ln 2 - \tfrac{2}{3}\pi^4\ln 2 - 3\pi^2\zeta(3) + \tfrac{1}{2}\pi^4\zeta(3) - \tfrac{5}{2}\pi^2\zeta(5) + \tfrac{\pi^6}{4} - \tfrac{3}{2}\pi^4 - \tfrac{\pi^8}{96}, \tag{34}$$

verified against the Hurwitz-tail anchor at 130 dps (residual 1.7×10^{-129}; `tests/test_smin_decomposition.py`). Two structural features are exact: $\zeta(7)$ cancels against the $\lambda(7)$ diagonal term, and all rational constants cancel to zero. Hence $S_{\min}$ lies in $\mathbb{Q}[\pi^2, \ln 2, \zeta(3), \zeta(5)]$: the two-loop constant carries *no irreducible depth-2 content at all* — every double t-value in its decomposition reduces to depth ≤ 1.

9.3 Inner photon sum: closed-form channel count

The inner photon sum $I(n_1, n_2) = \sum_{q\,\text{allowed}} W(n_1, n_2, q) \cdot d_q^T / \mu_q^p$ (at $p = 1$: $d_q^T/\mu_q = 1$) counts the total SO(4) CG-weighted vertex channels. Four structural properties:

(i) $I(0, n_2) = 0$ for all n_2 (structural zero: vertex parity at $n_{\text{ext}} = 0$ forces $2n_2$ odd, impossible).

(ii) $I(n_1, n_2) = I(n_2, n_1)$ (symmetric).

(iii) $I(n, n) = 2n$.

(iv) **Closed form**:

$$I(n_1, n_2) = 2\min(n_1, n_2), \qquad n_1, n_2 \geq 0. \tag{35}$$

Proof. Take $n_1 \leq n_2$ and $d = n_2 - n_1$. The window of Eq. (6) is $q \in \{d+1, d+3, \ldots, n_1 + n_2 - 1\}$, which has n_1 elements. For the component $((q+1)/2, (q-1)/2)$ the left triangle $((n_1+1)/2, (q+1)/2, (n_2+1)/2)$ holds for $d - 1 \leq q \leq n_1 + n_2 + 1$ and the right triangle $(n_1/2, (q-1)/2, n_2/2)$ for $d + 1 \leq q \leq n_1 + n_2 + 1$; for $((q-1)/2, (q+1)/2)$ the ranges are $d + 1 \leq q \leq n_1 + n_2 + 3$ and $d - 1 \leq q \leq n_1 + n_2 - 1$. All four integrality conditions are $n_1 + n_2 + q$ odd. Each component is therefore open on every window q, so $W = 2$ there and $I = 2n_1$. □ (Checked against `geovac/qed_vertex.py` for all $n_1, n_2 \leq 8$ in `tests/test_smin_decomposition.py`.)

9.4 Exact sunset relation

Substituting $I = 2\min$ into the double sum gives

$$S_{\text{sunset}} = 2\, S_{\min}, \tag{36}$$

where $S_{\text{sunset}} = \sum_{n_1, n_2 \geq 1} I(n_1, n_2)\, \phi(n_1)\, \phi(n_2)$ is the CG-exact two-loop sunset. Its value $S_{\text{sunset}} = 4.959873876068445108 82\ldots$ is verified at 35 dps, and the identity exactly in rational arithmetic at truncation $n \leq 30$, in `tests/test_smin_decomposition.py`. The two-loop sunset therefore has the closed form 2×Eq. (34) and carries no irreducible depth-2 content.

10 Three-Loop Results

10.1 Factorized $O(N^3)$ algorithm

The three-loop iterated sunset has chain topology: three electron lines (n_1, n_2, n_3) connected by two photon lines (q_1, q_2), with vertex selection at each junction. The chain topology permits factorization into partial sums:

$$S^{(3)} = \sum_{n_2=0}^{N} \frac{g_{n_2}}{|\lambda_{n_2}|^a} \cdot L(n_2) \cdot R(n_2), \tag{37}$$

where

$$L(n_2) = \sum_{\substack{n_1, q_1 \\ \text{allowed}}} \frac{W(n_1, n_2, q_1)\; g_{n_1}\; d_{q_1}^T}{|\lambda_{n_1}|^a\; \mu_{q_1}^p}, \qquad R(n_2) = \sum_{\substack{n_3, q_2 \\ \text{allowed}}} \frac{W(n_2, n_3, q_2)\; g_{n_3}\; d_{q_2}^T}{|\lambda_{n_3}|^a\; \mu_{q_2}^p}. \tag{38}$$

$W(n_1, n_2, q)$ is symmetric in (n_1, n_2), so $L = R$; the code computes them independently as a check. This reduces the five-fold sum $O(N^5)$ to $O(N^3)$ (each of L, R is $O(N^2)$ precomputed, then the outer sum is $O(N)$).

Verification: The factorized algorithm matches the direct $O(N^5)$ sum to $< 10^{-78}$ at $n_{\max} = 10, 15$ across four parameter combinations (default $a = 4, p = 1$; vertex-restricted; higher exponents $a = 6, p = 2$; with and without CG weights). The factorized algorithm at $n_{\max} = 50$ completes in < 60 seconds. Tests: 7 in `TestFactorizedMatchesDirect` and `TestFactorizedConvergence` (all passing; the frozen factorized-vs-direct gate is 10^{-25} — the $< 10^{-78}$ agreement quoted above is the measured value at the working precision, re-confirmed 2026-07-04).

10.2 Convergence analysis

At default exponents ($a = 4$, $p = 1$) with CG weights, the raw partial sums converge far too slowly for direct evaluation. An empirical power-law fit of the increments over $N \in [20, 120]$ gives

$$\Delta S^{(3)}(N) \approx 21.6 \cdot N^{-1.23}, \tag{39}$$

but the fitted exponent is an artifact: by the channel-count closed form (35) the inner partial sums grow as $L(n_2) = 4 \ln n_2 + O(1)$, so the increments follow $(\ln N)^2/N^2$ — a log-modulated law whose local log-slope rises from 0.9 to 1.4 over the fit window and masquerades as a power law. The true remaining tail at $N = 120$ is ≈ 12 units, i.e. *zero* reliable digits from raw partial sums (cf. Table 7: the $n_{\max} = 50$ partial sum is 21.58 against a limit of 40.66).

The reliable direct evaluation is a rigorous bracket with no series acceleration anywhere: the factorized sum has positive terms, so partial sums with the closed-form full-range inner sum give a strict lower bound, and the explicit inequalities $L(m) \le 4(\ln m + 1)$, $\phi(m) \le 2/(m+1)^2$ give an integral tail bound. At $N = 4 \times 10^6$,

$$S^{(3)} \in [40.65339,\ 40.65578] \tag{40}$$

(`tests/test_s3_decomposition.py` re-gates it at $N = 2 \times 10^4$). The decomposition assembly (Sec. 10.3) gives the canonical value, inside the bracket:

$$S^{(3)} = 40.65538\,75635\,85422\,54774\,43399\,69117\ldots \tag{41}$$

The unrestricted three-loop sum factorizes exactly as $D(a)^3$: at $a = 4$, $D(4)^3 = (\pi^2 - \pi^4/12)^3 \approx 5.379$, purely π^{even}.

10.3 Decomposition closure: the four trailing-1 identifications

Notation: $t_3(b_1, b_2, b_3) = \sum_{o_1 > o_2 > o_3 \ge 1 \text{ odd}} o_1^{-b_1} o_2^{-b_2} o_3^{-b_3}$, the Hoffman triple t-value in the same descending convention as $t(b, c)$. The channel-count closed form (35) gives $S^{(3)} = C - 12\,P\,S_{\min} - 9P^3$, with $P = \sum_{n \ge 1} \phi(n) = \pi^2 - \pi^4/12 - 64/81$ and C the three-chain core over odd $o = 2n + 3$; C expands into 12 triple t-values, 18 double t-values, and λ/boundary terms (exact in Fraction arithmetic at every truncation; `tests/test_s3_decomposition.py`).

The components with trailing argument $b_3 = 1$ carry an odd-harmonic factor $\widetilde{H}(j) \sim \frac{1}{2} \ln j$, and Levin-type series acceleration mis-converges on such log-modulated summands with an error that depends on the working precision. They are therefore evaluated by Abel summation over the harmonic weight, $t_3(b_1, b_2, 1) = \sum_{k \ge 1} (2k-1)^{-1} A(k)$ with $A(k) = \sum_{j \ge k} a_j$, where both sums are log-free; with these values the stuffle witnesses close at 5×10^{-216} and 8×10^{-217}. The bracket (40), which uses no acceleration, guards against the same failure mode.

PSLQ at 220/340 dps against the complete conjectural level-2 weight-homogeneous bases identifies the four components (residuals 2.6×10^{-216} to 7.3×10^{-339} in the archived dossier run; the frozen regression `tests/test_s3_decomposition.py` re-gates the identifications at 10^{-30}/45 dps):

$$t_3(2,1,1) = \tfrac{1}{2}\mathrm{Li}_4(\tfrac{1}{2}) + \tfrac{1}{48}\ln^4 2 + \tfrac{1}{24}\pi^2\ln^2 2 - \tfrac{19}{5760}\pi^4, \tag{42}$$

$$t_3(4,1,1) = \tfrac{1}{192}\pi^4\ln^2 2 - \tfrac{1}{64}\pi^2\ln 2\,\zeta(3) - \tfrac{11}{322560}\pi^6 - \tfrac{1}{2}\,t(5,1) - \tfrac{7}{128}\zeta(3)^2, \tag{43}$$

$$t_3(3,2,1) = \tfrac{3}{64}\pi^2\ln 2\,\zeta(3) - \tfrac{1}{9216}\pi^6 - \tfrac{1}{2}\,t(5,1) - \tfrac{49}{256}\zeta(3)^2, \tag{44}$$

$$\begin{aligned} t_3(3,4,1) = {} & \tfrac{1}{768}\pi^4\ln 2\,\zeta(3) + \tfrac{5}{128}\pi^2\ln 2\,\zeta(5) + \tfrac{61}{2903040}\pi^8 - \tfrac{1}{256}\pi^2\zeta(3)^2 \\ & + \tfrac{1}{2}\,t(5,3) - \tfrac{1}{2}\,t(7,1) - \tfrac{217}{512}\zeta(3)\zeta(5). \end{aligned} \tag{45}$$

Every component of the $S^{(3)}$ decomposition except the weight-10 graded piece ($W_{10} = -2048\,t_3(4,3,3) - 1024\,t_3(4,4,2) - 512\,t_2(8,2) - 1536\,t_2(4,6)$, treated below) is an identified element of the level-2 ring, with the genuine depth-2 generators $t(5,1)$ (weight 6) and $t(5,3)$, $t(7,1)$ (weight 8) entering at coefficients $\pm\frac{1}{2}$. **Realized motivic depth at three loops: ≤ 2 on every identified component** — one rung above $S_{\min}$'s depth ≤ 1, all catalogued, nothing new minted. Falsifier: `tests/test_s3_decomposition.py` (including the rigorous-bracket regression guard for the Levin/log failure mode).

Full assembly and the weight-10 block. Assembling the complete decomposition with all identified components yields a stronger statement than the entry-level reductions: *every depth-2 generator cancels in the total.* $t(5,1)$, $t(5,3)$, $t(7,1)$, and $\mathrm{Li}_4(\frac{1}{2})$ all carry total coefficient zero, as do $\zeta(7)$, $\zeta(9)$, $\zeta(11)$ and every rational constant — the $S^{(3)}$ echo of $S_{\min}$'s exact $\zeta(7)$ cancellation. The identified part of $S^{(3)}$ is a polynomial in π^2, $\ln 2$, $\zeta(3)$ and $\zeta(5)$ alone. The remaining weight-10 block reduces, by exact Fraction Gaussian elimination on the complete weight-10 quasi-shuffle closure (rank 157 on 255 words; identity verified numerically at 7×10^{-59}), to

$$\begin{aligned} W_{10}/256 = {} & 4\lambda(10) - \lambda(2)\lambda(8) - 3\lambda(4)\lambda(6) - 4\lambda(3)\,a(4,3) - 2\lambda(4)\,a(4,2) \\ & + a(8,2) + 4\,a(7,3) - 3\,a(6,4) - 4\,t_3(2,4,4) - 8\,t_3(3,3,4), \end{aligned} \tag{46}$$

where $a(b,c) := t(b,c) - t(c,b)$ and $\lambda(s) = (1-2^{-s})\zeta(s)$; the lower-weight $a(4,2)$, $a(4,3)$ are identified objects, leaving *five* surviving generators: the antisymmetric doubles $a(8,2)$, $a(7,3)$, $a(6,4)$ and the triples $t_3(2,4,4)$, $t_3(3,3,4)$. A structural obstruction protects this block from stuffle methods: the depth-≤ 3 relation content of the quasi-shuffle closure saturates already at the $\lambda\cdot t$ product stage, so no stuffle-product engineering at any depth can reduce it further.

The weight-10 block closed (2026-06-12). The missing input is the Charlton–Hoffman symmetry theorem [15] — level-2 regularized-double-shuffle content invisible to the stuffle-only system. Its generating-series form (their Thm. 2.21) at depth $m=3$, $\underline{\phi} = \underline{0}$, extracted at the coefficient of $y_1^{a-1}y_2^{b-1}y_3^{c-1}$ by exact-Fraction series algebra, gives the reversal-pair sums with every product term explicit (each instance used was verified numerically at 10^{-78} or better; `tests/test_s3_w10_identification.py`):

$$t_3(4,4,2) + t_3(2,4,4) = \tfrac{5}{6144}\pi^4\zeta(3)^2 - \tfrac{5}{256}\pi^2\zeta(3)\zeta(5) - \tfrac{1}{256}\pi^2\zeta(5,3) + \tfrac{137}{99532800}\pi^{10}, \tag{47}$$

$$t_3(4,3,3) + t_3(3,3,4) = \tfrac{37}{12288}\pi^4\zeta(3)^2 - \tfrac{25}{1024}\pi^2\zeta(3)\zeta(5) - \tfrac{889}{2048}\zeta(3)\zeta(7) - \tfrac{1}{256}\pi^2\zeta(5,3) + \tfrac{251}{58060800}\pi^{10}, \tag{48}$$

where $\zeta(5,3) = \sum_{m>n\geq 1} m^{-5}n^{-3}$ is the classical weight-8 depth-2 MZV generator (conjecturally irreducible over products of single zetas), entering through the weight-8 layer of the theorem's ζ-side product terms. The pair *differences* follow from four ordinary stuffles $t(x)\,t_2(y,z)$, $(x,y,z)\in$

$\{(2,4,4),(4,2,4),(3,3,4),(3,4,3)\}$, whose pairwise combinations eliminate the auxiliary triples $t_3(4,2,4)$, $t_3(3,4,3)$. Solving the two 2×2 systems identifies all four weight-10 triples individually:

$$t_3(2,4,4) = \tfrac{1}{4}a(8,2) - \tfrac{1}{4}a(6,4) - \tfrac{1}{6144}\pi^4\zeta(3)^2 - \tfrac{5}{512}\pi^2\zeta(3)\zeta(5) - \tfrac{1}{512}\pi^2\zeta(5,3) + \tfrac{373}{174182400}\pi^{10}, \quad (49)$$

$$t_3(3,3,4) = \tfrac{1}{4}a(7,3) - \tfrac{1}{4}a(6,4) + \tfrac{1}{12288}\pi^4\zeta(3)^2 + \tfrac{5}{1024}\pi^2\zeta(3)\zeta(5) - \tfrac{889}{4096}\zeta(3)\zeta(7) - \tfrac{1}{512}\pi^2\zeta(5,3) + \tfrac{251}{116121600}\pi^{10}, \quad (50)$$

each gated against its independently computed 200-digit value at residual $\leq 6\times 10^{-197}$ (likewise $t_3(4,4,2)$, $t_3(4,3,3)$).

Collapse theorem and the full closed form. Substituting Eqs. (49)–(50) into the full $S^{(3)}$ assembly, the three antisymmetric doubles $a(8,2)$, $a(7,3)$, $a(6,4)$ *cancel identically*: the explicit a-content of the weight-10 corollary is exactly absorbed by the a-content of the two triples (symbolic verification in the frozen falsifier). The weight-10 block therefore contributes

$$S^{(3)} - \text{Ident} \;=\; -7\pi^4\zeta(3)^2 + 70\pi^2\zeta(3)\zeta(5) + 6\pi^2\,\zeta(5,3) + \tfrac{179}{170100}\pi^{10}, \quad (51)$$

where Ident is the identified part of the symbolic assembly, and the complete closed form

$$S^{(3)} \;\in\; \mathbb{Q}\big[\pi^2,\ \ln 2,\ \zeta(3),\ \zeta(5),\ \zeta(5,3)\big] \quad (52)$$

is explicit:

$$\begin{aligned} S^{(3)} = {} & 128\pi^2\ln^2 2 - \tfrac{32}{3}\pi^4\ln^2 2 + 64\pi^2\ln 2 - \tfrac{304}{3}\pi^4\ln 2 + 16\pi^6\ln 2 - \tfrac{2}{3}\pi^8\ln 2 \\ & - 96\pi^2\ln 2\,\zeta(3) + 16\pi^4\ln 2\,\zeta(3) - 80\pi^2\ln 2\,\zeta(5) + 20\pi^2\zeta(3)^2 - 7\pi^4\zeta(3)^2 + 70\pi^2\zeta(3)\zeta(5) \\ & - 24\pi^2\zeta(3) + 40\pi^4\zeta(3) - 9\pi^6\zeta(3) + \tfrac{1}{2}\pi^8\zeta(3) - 20\pi^2\zeta(5) + 30\pi^4\zeta(5) - \tfrac{5}{2}\pi^6\zeta(5) \\ & + 6\pi^2\,\zeta(5,3) + \tfrac{8}{3}\pi^4 + \tfrac{374}{45}\pi^6 - \tfrac{8231}{3780}\pi^8 + \tfrac{124571}{680400}\pi^{10} - \tfrac{1}{192}\pi^{12}, \end{aligned} \quad (53)$$

with $\zeta(5,3)$ the *only* constant beyond the depth-≤ 1 ring, at coefficient $6\pi^2$. The closed form reproduces the canonical value (41); an independent route — the sum built on the earlier channel count, whose closed form was gated at 200 dps, plus the correction $2G + 4PS_{\min} - P^3 - 8PQ - R_3$ ($G = \sum \min(o_1,o_2)\,\phi(n_1)\,\phi(n_2)^2$ with $o_i = 2n_i+3$, $Q = \sum_n \phi(n)^2$, $R_3 = \sum_n \phi(n)^3$, all over $n \geq 1$) summed directly without the t-value table — agrees to 3×10^{-120} at 120 dps. Two structural readings: (i) the transient-generator cancellation pattern of the symbolic assembly ($t(5,1)$, $t(5,3)$, $t(7,1)$, $\mathrm{Li}_4(\frac{1}{2})$ all cancel) extends through weight 10 — the level-2 antisymmetric doubles $a(8,2)$, $a(7,3)$, $a(6,4)$ also carry total coefficient zero, and the *value* of $S^{(3)}$ is independent of these still-unreduced generators; (ii) the genuinely new depth-2 content at three loops enters on the *level-1* side ($\zeta(5,3) \in \mathcal{MT}(\mathbb{Z})$), while the level-2-specific content (the $\ln 2$ sector) stays at realized depth ≤ 1 — new depth arrives from level 1, new level stays at depth ≤ 1. Falsifiers: `tests/test_s3_w10_identification.py` (symmetry-pair sums, individual triples, symbolic a-collapse) and `tests/test_s3_decomposition.py` (closed form (53) and the independent route, with $\zeta(5,3)$ computed live).

Literature grounding (verified against the sources). All weight-≤ 7 closed forms in this section and Sec. 9.2 are verbatim rows of Hoffman's Appendix A [11] — including the triples $t_3(2,1,1)$, $t_3(4,1,1)$, $t_3(3,2,1)$, after eliminating $\zeta(\bar{5},1)$ via his own $t(5,1)$ row. The PSLQ identifications above therefore independently rediscovered the published table — a two-route verification.

The weight-≥ 8 reductions rest on the theorem-grade chain: Hoffman Cor. 4.1 (exact conversion to alternating MZVs) + parity/reduction theorems for alternating sums [12, 13] + the data mine's proven weight-≤ 12 reductions [14]. That $t_3(3,4,1)$ reduces to depth ≤ 2 is an instance of the depth-3 parity result for multiple t-values of Wang and Xu [16] (their Cor. 4.3: at weight 8 and depth 3 the parity combination is $2\,t_3$, which reduces to depth ≤ 2), so only the explicit coefficients of Eq. (45) are ours. We have not found those coefficients stated in [11], in the explicit families of [15] (whose t-value convention reverses the argument order), or in [16], whose only explicit depth-3 evaluations are at weight ≤ 6.

11 Depth-k Tower

Proposition 1 (Depth-k tower, corrected)**.** *At k loops on S^3 with vertex restrictions and CG weights:*

- *$k = 1$: free (unrestricted) integer-s traces of D^2 (equivalently $D(s)$ at even s) give π^{even} only (Theorem 1, proven); vertex-restricted quantities need not (see $k = 2$).*
- *$k = 2$ (vertex-parity sums): the two-loop sunset vertex restriction brings G and $\beta(4)$ ($\beta(4)$ first enters here among the identified quantities; the χ_{-4} theorem).*
- *$k = 2$ (CG-weighted): $S_{\text{sunset}} = 2S_{\min}$ is an identified element of $\mathcal{MT}(\mathbb{Z}[1/2])$ at level 2. The depth is* bounded *by 2 (the nesting depth of the min-weight double sum) but the content decomposes into Hoffman double t-values, which are classical period-ring elements (Sec. 9.2).*
- *$k = 3$: $S^{(3)}$ is a fully identified element of $\mathbb{Q}[\pi^2, \ln 2, \zeta(3), \zeta(5), \zeta(5,3)]$ with* realized *depth ≤ 2 unconditionally (Eq. (52)): the sole depth-2 constant is the level-1 generator $\zeta(5,3)$, at coefficient $6\pi^2$; every level-2 depth-2 generator ($t(5,1)$, $t(5,3)$, $t(7,1)$, $a(8,2)$, $a(7,3)$, $a(6,4)$) appears transiently and cancels in the total (Sec. 10.3).*
- *$k = 4$: the* realized *depth of $S^{(4)}$ is ≤ 3. Depth-4 content enters $S^{(4)}$ only through the four-chain core C_4 (the o-space relation $S^{(4)} = C_4 - 6PC - 3S_o^2 + 27S_oP^2 - 27P^4$, with $S_o = \sum \min(o_1, o_2)\,\phi(n_1)\phi(n_2)$, gives C_4 coefficient $+1$; the other cores are $\leq$ depth-3;* `tests/test_s4_stage1.py`*), so the verdict reduces to whether C_4's depth-4 part reduces to depth ≤ 3. At weights $9, 11, 13$ the quasi-shuffle (stuffle) relation system spans the* entire *depth-4 word space (rank = number of depth-4 words: $35, 84, 165$; exact rank over two independent primes), collapsing C_4's depth-4 content to depth ≤ 3 unconditionally. At weights $5, 7$ the stuffle system is rank-deficient, but these weights lie below the threshold for any depth-≥ 3 irreducible: weight-5 values reduce to single polylog constants (depth ≤ 1) and weight-7 to depth ≤ 2 (the $w \leq 7$ GeoVac multiple-t values match Hoffman's Appendix-A reductions, verified previously). The explicit closed form of $S^{(4)}$ remains open; the depth verdict does not depend on it.*

Loop order bounds the depth of the classical level-2 multiple-t-value content; it does not mint new irreducible transcendentals automatically. Realized depth so far grows strictly more slowly than loop order ($\leq k - 1$ at $k = 2, 3, 4$).

The mechanism: each vertex coupling creates one level of Hurwitz nesting. One vertex produces a depth-2 double t-value sum ($S_{\min}$). Two vertices produce a depth-≤ 3 triple t-value sum ($S^{(3)}$). The prediction extends: k vertices produce a depth-$\leq (k+1)$ multiple t-value in $\mathcal{MT}(\mathbb{Z}[1/2])$. Whether the surviving irreducible depth-k component is new (not reducible by parity/stuffle within

the level-2 algebra) is a question for each k separately — the structural nesting is genuine, but the motivic content is classical at $k = 2$ and, on all identified components, at $k = 3$ (realized depth ≤ 2). This is the S^3 analog of the increasing motivic depth observed in flat-space multi-loop Feynman integrals [8, 10].

12 Fine-Structure Euler Sum Cancellation

The Dirac Dirichlet series $D(s)$ studied in the preceding sections sums over the *free* Camporesi–Higuchi spectrum with uniform degeneracy weights g_n. A complementary probe arises from the *exact* Dirac hydrogen spectrum, where the degeneracy-weighted energy expansion in powers of $(Z\alpha)^2$ generates Dirichlet sums with state-dependent coefficients that mix harmonic numbers of varying depth.

12.1 Setup

The exact Dirac energy for quantum numbers (n, j) is

$$E_{nj} = mc^2\left(1 + \frac{(Z\alpha)^2}{\left(n_r + \sqrt{\kappa^2 - (Z\alpha)^2}\right)^2}\right)^{-1/2} - mc^2\,, \tag{54}$$

where $\kappa = j + \frac{1}{2}$ and $n_r = n - \kappa$. Expanding in powers of $x = (Z\alpha)^2$,

$$\sum_j d(n,j)\,\frac{E_{nj}}{mc^2} = \sum_{p=1}^{\infty} c_p(n)\,x^p\,, \tag{55}$$

where $d(n, j) = 2(2j+1)$ for $j < n - \frac{1}{2}$ and $d(n, j) = 2j+1$ for $j = n - \frac{1}{2}$. The degeneracy-weighted Dirichlet sums are

$$D_p \;=\; \sum_{n=1}^{\infty} c_p(n)\,. \tag{56}$$

12.2 Closed forms for $c_p(n)$

Symbolic expansion of the exact Dirac formula and summation over j gives closed forms involving generalized harmonic numbers $H_{n-1}^{(r)} = \sum_{k=1}^{n-1} k^{-r}$:

$$c_2(n) = -\frac{5n-4}{4n^3}\,, \tag{57}$$

$$c_3(n) = \frac{19n-20}{8n^5} - \frac{3}{2}\,\frac{H_{n-1}}{n^4} - \frac{1}{2}\,\frac{H_{n-1}^{(2)}}{n^3}\,. \tag{58}$$

At order $p = 2$ and $p = 3$, the sums D_2 and D_3 collapse to pure single-zeta values:

$$D_2 = -\frac{5}{4}\,\zeta(2) + \zeta(3)\,, \qquad D_3 = \frac{19}{8}\,\zeta(4) - \frac{11}{4}\,\zeta(5)\,. \tag{59}$$

12.3 $\zeta(3)$ complementarity

The $p = 2$ coefficient defines the *fine-structure shell sum*

$$\mathcal{S}(n) \;=\; \sum_j d(n,j)\Big(\frac{n}{j+\frac{1}{2}} - \frac{3}{4}\Big) \;=\; \frac{n(5n-4)}{2}\,, \tag{60}$$

so that $c_2(n) = -\mathcal{S}(n)/(2n^4)$. It splits as $-3n^2/2 + 2n(2n-1)$, the second piece twice a hexagonal number.

In the Fock convention ($n \geq 1$), the single-chirality Dirac mode degeneracy is $g_n^{\text{Dirac}} = 2n(n+1)$.

Theorem 5 ($\zeta(3)$ complementarity)**.** *For all $n \geq 1$,*

$$\mathcal{S}(n) + g_n^{\text{Dirac}} \;=\; \frac{9n^2}{2} \;=\; |\lambda_0|^2 \times n^2_{\text{Schr}}\,, \tag{61}$$

where $|\lambda_0| = 3/2$ is the lowest Dirac eigenvalue on S^3 and $n^2_{\text{Schr}} = 2n^2$ is the Schrödinger degeneracy (including spin). At the Dirichlet series level,

$$D_{\mathcal{S}}(s) + D_g(s) \;=\; \frac{9}{2}\,\zeta(s{-}2) \qquad \textit{for all } s\,, \tag{62}$$

where $D_{\mathcal{S}}(s) = \sum_{n\geq 1} \mathcal{S}(n)\, n^{-s}$ and $D_g(s) = \sum_{n\geq 1} g_n^{\text{Dirac}}\, n^{-s}$. The $\zeta(s{-}1)$ terms cancel identically.

Proof. $\mathcal{S}(n) = 5n^2/2 - 2n$ and $g_n^{\text{Dirac}} = 2n^2 + 2n$; their sum is $9n^2/2$. Therefore $D_{\mathcal{S}}(s) = \frac{5}{2}\,\zeta(s{-}2) - 2\,\zeta(s{-}1)$ and $D_g(s) = 2\,\zeta(s{-}2) + 2\,\zeta(s{-}1)$; summing gives $\frac{9}{2}\,\zeta(s{-}2)$. □

The period content is in the cancelled term: at odd s, $\zeta(s{-}1)$ is an even zeta value, but at even s it is an odd one ($\zeta(3)$, $\zeta(5)$, …). At $s = 4$:

$$D_{\mathcal{S}}(4) = \frac{5}{2}\,\frac{\pi^2}{6} - 2\zeta(3), \tag{63}$$

so the $-2n$ piece of $\mathcal{S}$ carries $\zeta(3)$, with sign opposite to the $+2\zeta(3)$ in $D_g(4) = 2\zeta(2) + 2\zeta(3)$. Their sum $\frac{9}{2}\zeta(2) = 3\pi^2/4$ lies in $\mathbb{Q}[\pi^2]$. This is an analogue, in the fine-structure sector, of T9's integer-s statement (Theorem 1), not T9 itself, which concerns the half-integer spectrum $n + 3/2$; the complementarity is a sum over integer n with $g = 2n(n+1)$.

12.4 D_4 at weight 7: product zeta survives

At order $p = 4$,

$$\begin{aligned} c_4(n) = &-\frac{5(41n-40)}{64n^7} + \frac{15}{4}\,\frac{H_{n-1}}{n^6} - \frac{1}{4}\,\frac{H^{(2)}_{n-1}}{n^5} \\ &- \frac{3}{4}\,\frac{H^{(3)}_{n-1}}{n^4} - \frac{1}{4}\,\frac{H^{(4)}_{n-1}}{n^3}\,. \end{aligned} \tag{64}$$

The harmonic-number sums decompose into weight-7 Euler sums $S_{r,s} = \sum_{n\geq 1} H_n^{(r)}/n^s$ via the identity $H^{(r)}_{n-1} = \zeta(r) - \zeta(r,n)$ and Euler's formula $S_{1,6} = 4\zeta(7) - \zeta(2)\zeta(5) - \zeta(3)\zeta(4)$ [17].

Observation 1 (Euler sum cancellation at weight 7)**.** *At 200 dps, PSLQ identifies*

$$D_4 = -\frac{205}{64}\,\zeta(6) + \frac{71}{8}\,\zeta(7) - \frac{9}{2}\,\zeta(3)\zeta(4)\,, \tag{65}$$

with residual $< 10^{-195}$ *(archived PSLQ run; the value is pinned live in* `tests/test_paper28_gap_closure_pins.py`*, and the* D_4/D_5 *decimals were independently cross-verified by direct partial summation, 2026-07-04). The product* $\zeta(2)\zeta(5)$ *cancels* algebraically*: its coefficient vanishes exactly (verified in exact rational arithmetic from the Euler sum decomposition), while* $\zeta(3)\zeta(4)$ *survives with coefficient* $-9/2$*.*

This breaks the pure-single-zeta pattern: D_2 and D_3 involve only $\zeta(s)$ values, whereas D_4 introduces the first zeta product. The four weight-7 Euler sums, independently identified by PSLQ, are

Sum	$\zeta(7)$	$\zeta(2)\zeta(5)$	$\zeta(3)\zeta(4)$
$S_{1,6}$	4	-1	-1
$S_{2,5}$	-10	5	2
$S_{3,4}$	18	-10	0
$S_{4,3}$	-17	10	1
$S_{5,2}$	11	-4	-2

The stuffle relations $S_{r,s} + S_{s,r} = \zeta(r)\zeta(s) + \zeta(r+s)$ are satisfied to $< 10^{-200}$.

12.5 D_5 at weight 9: product survival rule

At order $p = 5$,

$$\begin{aligned} c_5(n) = {} & \frac{7(71n-72)}{128\,n^9} - \frac{105}{16}\,\frac{H_{n-1}}{n^8} + \frac{45}{16}\,\frac{H^{(2)}_{n-1}}{n^7} + \frac{5}{4}\,\frac{H^{(3)}_{n-1}}{n^6} \\ & - \frac{3}{8}\,\frac{H^{(4)}_{n-1}}{n^5} - \frac{15}{32}\,\frac{H^{(5)}_{n-1}}{n^4} - \frac{5}{32}\,\frac{H^{(6)}_{n-1}}{n^3}\,, \end{aligned} \tag{66}$$

verified as exact rational equality against direct sympy expansion for $n = 1, \ldots, 15$. The Dirichlet sum decomposes via weight-9 Euler sums $S_{r,s}$ for $r + s = 9$ (computed via Hurwitz zeta to 250 dps, stuffle relations verified to $< 10^{-200}$):

Sum	$\zeta(9)$	$\zeta(2)\zeta(7)$	$\zeta(3)\zeta(6)$	$\zeta(4)\zeta(5)$
$S_{2,7}$	$-35/2$	7	2	4
$S_{3,6}$	$85/2$	-21	0	-6
$S_{4,5}$	$-125/2$	35	0	5
$S_{5,4}$	$127/2$	-35	0	-4
$S_{6,3}$	$-83/2$	21	1	6
$S_{7,2}$	$37/2$	-6	-2	-4

PSLQ at 200 dps identifies

$$D_5 = \frac{497}{128}\,\zeta(8) - \frac{467}{16}\,\zeta(9) + \frac{385}{32}\,\zeta(3)\zeta(6) + \frac{75}{8}\,\zeta(4)\zeta(5)\,, \tag{67}$$

with residual $< 10^{-190}$. The product $\zeta(2)\zeta(7)$ cancels algebraically (verified from the Euler sum collection), while both $\zeta(3)\zeta(6)$ and $\zeta(4)\zeta(5)$ survive. This extends the pattern from D_4: product terms grow from 0 (at D_2, D_3) to 1 (at D_4) to 2 (at D_5).

12.6 D_6 at weight 11: product survival rule extended

At order $p = 6$ the weight-11 Euler sums $S_{r,s}$ for $r + s = 11$ are computed via Hurwitz zeta to 400 dps, with stuffle relations verified to $< 10^{-300}$. Seven nontrivial sums ($r = 1, \ldots, 8$, $r \neq 3$) contribute through the β_r coefficients of the $c_6(n)$ expansion. Individual decompositions are found by PSLQ against the weight-11 depth-1 MZV basis $\{\zeta(11), \zeta(2)\zeta(9), \zeta(3)\zeta(8), \zeta(4)\zeta(7), \zeta(5)\zeta(6)\}$:

Sum	$\zeta(11)$	$\zeta(2)\zeta(9)$	$\zeta(3)\zeta(8)$	$\zeta(4)\zeta(7)$	$\zeta(5)\zeta(6)$
$S_{1,10}$	6	-1	-1	-1	-1
$S_{2,9}$	-27	9	2	6	4
$S_{4,7}$	$-329/2$	84	0	21	4
$S_{5,6}$	$463/2$	-126	0	-21	0
$S_{6,5}$	$-461/2$	126	0	21	1
$S_{7,4}$	$331/2$	-84	0	-20	-4
$S_{8,3}$	-82	36	1	15	6

The $\zeta(3)\zeta(8)$ column has a sparse support pattern: nonzero only for $S_{1,10}$, $S_{2,9}$, and $S_{8,3}$.

Collecting through the β_r coefficients and the rational $\zeta(10)$, $\zeta(11)$ parts:

$$D_6 = -\frac{2289}{512}\,\zeta(10) + \frac{1589}{16}\,\zeta(11) - \frac{1617}{64}\,\zeta(3)\zeta(8) - \frac{1785}{64}\,\zeta(4)\zeta(7) - \frac{2065}{64}\,\zeta(5)\zeta(6)\,, \tag{68}$$

verified to 62 digits against direct Euler-sum evaluation at 400 dps. The product $\zeta(2)\zeta(9)$ cancels algebraically, and all three surviving products $\zeta(3)\zeta(8)$, $\zeta(4)\zeta(7)$, $\zeta(5)\zeta(6)$ are present, matching the prediction $\max(0, \lfloor(2 \cdot 6 - 5)/2\rfloor) = 3$.

Observation 2 (Product survival rule)**.** *At each Sommerfeld order $p \geq 2$, the Dirichlet sum D_p decomposes into weight-$(2p-1)$ zeta values. The product $\zeta(2)\zeta(2p-3)$ cancels algebraically at every order. Exactly* $\max(0, \lfloor(2p-5)/2\rfloor)$ *product terms survive, all of the form $\zeta(a)\zeta(b)$ with $a+b = 2p-1$, a odd ≥ 3, b even ≥ 4.*

Verified through D_6 ($p = 6$, weight 11); no derivation of the rule is known.

12.7 Transcendental structure

Substituting $\zeta(6) = \pi^6/945$ and $\zeta(4) = \pi^4/90$, equation (65) becomes

$$D_4 = -\frac{41\pi^6}{12096} + \frac{71}{8}\,\zeta(7) - \frac{\pi^4}{20}\,\zeta(3)\,. \tag{69}$$

This is a three-tier decomposition: π^6 (weight-6 even-zeta), $\zeta(7)$ (weight-7 odd-zeta), and the mixed product $\pi^4 \cdot \zeta(3)$ (weight-4 even $\times$ weight-3 odd). In Paper 18's operator-order taxonomy, π^{even} comes from Δ_{LB} and D^2 (not from every second-order operator; Theorem 1's scope) and $\zeta(\text{odd})$ from first-order operators; the mixed product $\pi^4 \cdot \zeta(3)$ has both tiers multiplicatively in a single observable.

The $\zeta(2)\zeta(5)$ cancellation is non-trivial: four of the five contributing Euler sums carry nonzero $\zeta(2)\zeta(5)$ coefficients ($S_{1,6}$, $S_{2,5}$, $S_{4,3}$, $S_{5,2}$), and the cancellation emerges only in the specific linear combination that the Dirac degeneracy weights and harmonic-number structure select. In contrast, $S_{3,4}$ alone is free of $\zeta(3)\zeta(4)$, but the remaining four sums contribute $\zeta(3)\zeta(4)$ terms that do not cancel.

Table 5: Even/odd decomposition of $D(s)$: PSLQ at 80+ dps.

s	$D_{\text{even}}(s)$	$D_{\text{odd}}(s)$	$D_{\text{even}} - D_{\text{odd}}$
4	$\frac{\pi^2}{2} - \frac{\pi^4}{24} - 4G + 4\beta(4)$	$\frac{\pi^2}{2} - \frac{\pi^4}{24} + 4G - 4\beta(4)$	$8(\beta(4) - G)$
5	odd-zeta + Dirichlet	odd-zeta − Dirichlet	$16(\beta(5) - \beta(3))$
6	π^{even} + Dirichlet	π^{even} − Dirichlet	$32(\beta(6) - \beta(4))$

Table 6: Verification of Theorem 3: $D_{\text{even}}(s) - D_{\text{odd}}(s) = 2^{s-1}(\beta(s) - \beta(s-2))$.

s	2^{s-1}	Residual $\lvert\text{LHS} - \text{RHS}\rvert$
2	2	0 (sympy)
3	4	0 (sympy)
4	8	$< 10^{-100}$
5	16	$< 10^{-100}$
6	32	$< 10^{-100}$
7	64	$< 10^{-100}$
8	128	$< 10^{-100}$
9	256	$< 10^{-100}$
10	512	$< 10^{-100}$
15	16384	$< 10^{-100}$

13 Comprehensive Transcendental Tables

14 Discussion

The periods met in this paper sort into four layers. Single spectral sums of D^2 at integer s lie in $\mathbb{Q}[\pi^2]$ (Theorem 1); single sums of $|D|$ at odd $s \geq 5$ are rational combinations of two odd zeta values (Theorem 2); restricting n to one parity class adds the level-4 values G and $\beta(4)$, which cancel in the full sum (Theorem 3). Nested chain sums lie in the level-2 (alternating) multiple zeta values with realized depth $\leq k-1$ at loop order $k = 2, 3, 4$ (Proposition 1). At two loops every double t-value reduces to depth ≤ 1. At three loops the closed form (Eq. (53)) has one depth-2 constant, $\zeta(5,3)$, which is a level-1 period ($\zeta(5,3) \in \mathcal{MT}(\mathbb{Z})$), while every level-2 depth-2 generator met along the way cancels: new depth arrives from level 1, and the $\ln 2$ sector stays at depth ≤ 1. Sums over the exact Dirac hydrogen spectrum give weight-$(2p-1)$ Euler sums with a product-cancellation pattern observed through $p = 6$ (Observation 2).

The vertex model enters the nested sums only through its window. Because $W = 2$ on every window triple and 0 elsewhere (Sec. 9.3), the channel-count weight multiplies each vertex-restricted sum by a power of 2, so the period content above is a property of the window $|n_1 - n_2| \leq q \leq n_1 + n_2$, $n_1 + n_2 + q$ odd. The physical vector-photon vertex has a different window and nonconstant reduced amplitudes (Paper 33); the periods of the corresponding sums are not determined here.

The depth-k tower fits the framework of Zagier's weight conjecture [9]: loop order raises the weight, and the algebra of multiple zeta values receives the result. The half-integer shift $a = 3/2$ of the Dirac spectrum is what puts the sums in the odd-denominator (t-value) form of the level-2 ring.

Table 7: Three-loop factorized convergence at $a = 4$, $p = 1$, CG-weighted. The limit is $S^{(3)} = 40.6554\ldots$ (Eq. (41)): raw partial sums carry zero reliable digits at these truncations — the $(\ln N)^2/N^2$ increment law, not the naively fitted power law, governs the tail (Sec. 10). Rows computed from `geovac/qed_three_loop.py`, checked against an independent exact-rational route, and pinned in `tests/test_paper28_gap_closure_pins.py`.

$n_{\max}$	$S^{(3)}(n_{\max})$	Δ from previous	Time (s)
5	2.753	—	< 0.1
10	6.836	4.083	< 0.1
15	10.181	3.345	0.2
20	12.837	2.655	0.5
50	21.581	—	< 60
∞	$40.6554\ldots$	—	(decomposition route)

14.1 Open questions

1. Does the depth-k tower stabilize? Realized depth is $\leq k-1$ at $k = 2, 3, 4$; the explicit closed form of $S^{(4)}$ and the continuation to $k \geq 5$ are open.

2. Is there a Picard–Fuchs interpretation of $S_{\min}$, analogous to Apéry's recurrence for $\zeta(3)$ [18]?

3. What are the periods of the chain sums built on the physical vector-photon vertex, with its window and its reduced amplitudes (Paper 33)? This needs the normalization of that vertex.

4. Can the product survival rule (Observation 2) be derived for all p from Euler-sum reflection identities?

5. The model's magnetic-moment ratio: its normalization is owed (Sec. 7). Within the model, is the asymptotic exponent of Eq. (22) exactly -2; can the prefactor in $V_{\mathrm{mag}} \to (4/3)/|\lambda|$ be derived in closed form; and can the telescoping recurrence of the exact square-root forms of V_{mag} and $B(n_{\mathrm{int}} = 1)$ (the negative square-root argument at level n equals the positive one at level $n-1$) be proven from the coupling rules?

References

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